

MADHAV KUMAR

Software Engineer | Backend & Cloud

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PROFILE

Software Engineering intern at Nielsen with hands-on experience in building and optimizing large-scale distributed systems, improving system reliability, and automating production workflows using AWS, Python, and Java. Strong in backend development, cloud infrastructure, and CI/CD-driven deployments.

TECHNICAL SKILLS

Core Skills:

Python, Java, AWS (Lambda, S3, EC2, RDS, DynamoDB, SQS, SNS, CloudWatch, IAM), Docker, Kubernetes, Terraform, PostgreSQL, ClickHouse, GitLab CI/CD, Git

Additional Experience:

Spring Boot, MySQL, MongoDB, REST APIs, Microservices, Linux, Jira, Confluence

EXPERIENCE

Nielsen – Software Development Engineer Intern

Mar 2025 - Present

- Resolved critical audio capture failures (**2000+ daily runs**) by diagnosing **cross-layer sampling mismatches**, reducing audio dropouts from **20–25% to <1%** and **restoring SLA compliance**.
- Stabilized audio processing pipeline by standardizing end-to-end configuration and **optimizing buffer size (42.6 ms)**, eliminating signal loss during capture.
- Automated fleet-wide configuration rollout across **450+ multi-region Windows VMs** using **PowerShell scripts** over VPN, ensuring consistent deployment and system stability.
- Engineered an **AWS Lambda validation framework** using Python and CloudWatch, reducing manual log analysis effort by **60%** and improving deployment reliability.
- Enabled automated **UAT validation** within **1 CI/CD pipeline**, **improving release reliability** and eliminating manual validation steps.
- Enhanced media processing workflows by implementing **WAV-to-FLAC compression (70% size reduction)** and building **event-driven pipelines (PostgreSQL, SNS, SQS, ClickHouse)** for scalable asset tracking and validation.

PROJECTS

GitForgeX — Custom Version Control System

- Built a **Git-inspired Version Control System** using locally running **containerized services**, including a Repository Management Service and a Version History Service.
- Developed secure handshake-based push operations, **API-driven inter-service** communication, and snapshot-based repository storage with low-latency handling of multiple requests.

Kubernetes Logging System (Personal Project)

- Built a **Kubernetes-based sidecar system** using **ClickHouse and Fluentd** for log collection and analysis.
- Successfully collected and queried over **50,000** log events in local tests with sub-second response times.

Sentiment Analysis API (Spring Boot + Java NLP)

- Developed a Spring Boot application that analyzes user input text using **Stanford CoreNLP** to detect sentiment.
- Designed a simple REST API for real-time **sentiment classification** and response delivery.

Task Scheduler App (Android – Kotlin + SQLite)

- Created a fully offline **Android task manager** with support for recurring tasks, categories, and instant search.
- Tested with **2,000+ tasks** on mid-range devices while maintaining smooth performance and **< 50 ms** search response.

CERTIFICATES

Application Development in Java

- Gained practical knowledge in Java GUI development, JDBC for database management, and Java networking. Applied these skills in projects to improve application design and functionality.

EDUCATION

Lovely Professional University

Bachelor of Technology-Computer Science & Engineering - 87.60%

Jalandhar, Punjab, India

2022 - 2026

Litera Valley School

Senior secondary education (PCM 11th - 12th) - 81%

Patna, Bihar, India

2019 - 2021

PTJM Saraswati Vidya Mandir

Secondary education (10th) - 92%

Rajgir, Bihar, India

2018 - 2019